# Patrick Thangarajah

Product Design | Prototyping | Testing & Validation | Simulation

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**Hiring Team** *12 November 2026*

DarkVision Technologies Inc.
Calgary, AB

**Mechanical Test Engineer**

Dear DarkVision Hiring Team,

I recently completed my MASc in Mechanical and Mechatronics Engineering at the University of Waterloo and am currently based in Calgary. I'm interested in DarkVision's Mechanical Test Engineer position because of the opportunity to design test equipment and evaluate how well devices perform under different conditions. Through my projects and work experience, I've enjoyed the practical side of engineering, especially building test systems and troubleshooting problems.

I've gained much of my practical design experience through building test equipment. During my UBC capstone, I developed silicone and agar bladder phantoms and built the control and data acquisition interface for a pressurised test system. The system monitored pressure and volume in real time and performed automated fill and void cycles. I validated stable, repeatable pressurisation at 2.2 kPa and tested up to 4.75 kPa. I continued working on the project with Echoscope, where I designed and iterated the PCB and wrote five manufacturing and verification procedures that were approved and used by the client.

During my graduate research, I validated an LS-DYNA model against physical sled tests and investigated differences in positioning, restraint fit and response timing. This involved understanding why the simulations and tests produced different results, which is something I'd like to apply to investigating test failures at DarkVision. During my internship at GN Corporation in Airdrie, I prepared work instructions and SOPs based on engineering drawings and customer specifications. I also reduced fibreglass cutting waste by 50% through process analysis. I'm comfortable using SolidWorks, 3D printing, a lathe, mill and drill press.

I'm a Canadian citizen with a Class 5 driver's licence and am happy to split my time between Calgary and Edmonton. I'm available to start immediately and would appreciate the opportunity to discuss the position further and how my experience could contribute to the team.

Sincerely,

**Patrick Thangarajah**